Reg. No. | | | | | | | | | | | [12ME752]

SEVENTH SEMESTER B.TECH EXAMINATION DECEMBER-2015

COMPUTATIONAL FLUID DYNAMICS

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full Questions.

1. a. Compare the analytical, experimental and computational approaches of solving a fluid mechanics and Heat transfer problem. **05**

b. Write the steady state, steady flow energy equation. Mention the various terms of this equation. How this equation is reduced to the form of Bernoulli's equation? **05**

c. In a three-dimensional incompressible flow, the velocity components in y and z directions are $v = ax^3 - by^2 + cz,\ w = bx^3 - cy^2 + az^2x$. Determine the missing component of velocity such that the continuity equation is satisfied. **04**

2. a. Explain the physical classification of partial differential equations. **05**

b. Find the real root of the equation $Cosx = 3x - 1$ using the method of iterations. **05**

c. Find the real root of $xe^x - 2 = 0$ using Newton – Raphson method. **04**

3. a. Solve the following system of equations using Gauss-seidal iteration method. **06**

$$10x + 2y + z = 9$$
$$x + 10y - z = -22$$
$$-2x + 3y + 10z = 22$$

b. Solve the following set of simultaneous equations by Crout's method: **08**

$$3x + y + z = 8$$
$$x + 2y + 2z = 11$$
$$2x + y + 3z = 13$$

4. a. Given $\frac{dy}{dx} = \frac{1}{x+y}$ with $y(0) = 2$. If $y(0.2) = 2.09, y(0.4) = 2.17, y(0.6) = 2.24$, find $y(0.8)$ using Milne's Predictor-Corrector formula. **06**

b. Find $y(0.1)$ and $y(0.2)$ using Runge-Kutta 4th order method given that $\frac{dy}{dx} = y - \frac{2x}{y}$ take y(0)=1 **08**

5. a. Differentiate between truncation error, discretization error and round of error in a Numerical scheme. **06**

b. Consider the steady state heat conduction in a slab of thickness L in which energy **08**

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is generated at a constant rate of $Q_g = 7.2\times10^7 W/m^3$. The boundary surface at x=0 is maintained at a constant temperature T_o while the boundary surface at x=L dissipates heat by convection with heat transfer coefficient h into an ambient at temperature T_∞. Dividing the region into three equal sub regions, write the finite difference equations of this heat conduction problem. Compute the temperatures at the nodes. Take

$h = 200W/m^2K, K = 18W/mK, L = 1.2cm, T_\infty = 120^oC, T_o = 60^oC.$

6. **a.** Consider the steady state heat conduction in a rectangular region $0 \le x \le 3b, 0 \le y \le 2b$ subjected to the boundary conditions shown in Fig.6 (a). Formulate the temperature distribution equations at six nodes show in the figure and express these equations in the matrix form. **10**

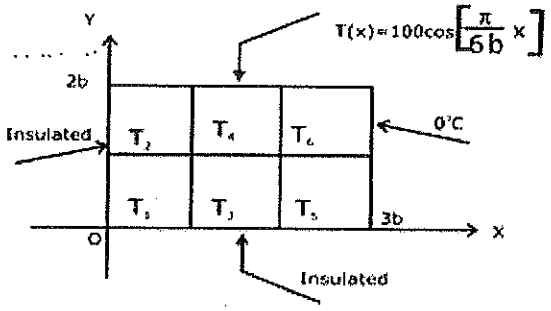

Fig-6(a)

b. A plane wall of thickness 2L with constant thermal properties is initially at a uniform temperature T_i. Suddenly one of the surfaces is subjected to heating by the flow of hot gases at temperature T_∞ with heat transfer coefficient h. The other surface is kept insulated; write the mathematical formulation of this heat conduction problem. **04**

7. **a.** Write the generic form of general transport equation and deduce it to get mass, momentum and energy equation. **06**

b. Consider the steady one-dimensional convection diffusion equation of a property Φ **08**

$$\frac{d}{dx}(\rho u \phi) = \frac{d}{dx}\left(\Omega \frac{d\phi}{dx}\right).$$

Using control volume approach, discretize this equation and obtain the neighboring co-efficients using central difference scheme.

8. Write short notes on:-

i. Advances in computational methods. **06**

ii. CFD solutions for multiphase flow and compressible flow problems. **08**